THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS CHAPTER TEST SET A

Chapter 7: **The Mathematics of Maybe: Introduction to Probability** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A fair die is rolled once. The probability of getting a prime number is: 1

(a) $\frac{1}{3}$ **(b)** $\frac{2}{3}$ **(c)** $\frac{1}{2}$ **(d)** $\frac{1}{4}$

2. A fair die is rolled once. The probability of getting a number less than 3 is: 1

(a) $\frac{1}{2}$ **(b)** $\frac{1}{3}$ **(c)** $\frac{2}{3}$ **(d)** $\frac{1}{4}$

3. A bag contains 5 red, 3 blue and 2 green balls. One ball is drawn at random. The probability that it is blue is: 1

(a) $\frac{3}{10}$ **(b)** $\frac{7}{10}$ **(c)** $\frac{2}{5}$ **(d)** $\frac{3}{11}$

4. A bag contains 6 red, 4 blue and 5 green balls. One ball is drawn at random. The probability that it is red is: 1

(a) $\frac{3}{5}$ **(b)** $\frac{1}{5}$ **(c)** $\frac{1}{3}$ **(d)** $\frac{2}{5}$

5. A letter is picked at random from the word ‘PROBABILITY’. The probability of picking the letter B is: 1

(a) $\frac{1}{6}$ **(b)** $\frac{2}{11}$ **(c)** $\frac{9}{11}$ **(d)** $\frac{3}{11}$

6. A letter is picked at random from the word ‘MATHEMATICS’. The probability of picking a vowel is: 1

(a) $\frac{7}{11}$ **(b)** $\frac{4}{11}$ **(c)** $\frac{1}{3}$ **(d)** $\frac{5}{11}$

7. An event occurred 10 times in 25 trials. Its experimental probability (relative frequency) is: 1

(a) 2.5 **(b)** 0.5 **(c)** 0.4 **(d)** 0.2

8. The number of outcomes in the sample space when two dice are rolled together is: 1

(a) 38 **(b)** 36 **(c)** 72 **(d)** 18

9. The probability of any event always lies between: 1

(a) 0 and 1 (both included) **(b)** 0 and 100

(c) 1 and 2 **(d)** −1 and 1

10. A fair coin has landed heads six times in a row. The probability of tails on the next toss is: 1

(a) $\frac{1}{3}$ **(b)** $\frac{3}{4}$ **(c)** $\frac{1}{2}$ **(d)** $\frac{1}{64}$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Write the sample space when two coins are tossed together and find the probability of exactly one head. 2

12. A fair die is rolled. Find the probability of getting a prime number. 2

13. A bag has 4 red and 6 blue balls. A ball is drawn at random. What is the probability that it is not red? 2

14. A die is rolled 50 times and shows a 4 eight times. Find the experimental probability of a 4 and compare it with the theoretical probability. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Three coins are tossed together. Write the sample space and find the probability of at least two heads. **3**

16. Two dice are rolled. Find the probability that **3**

(i) the sum is 8

(ii) the sum is at least 10.

17. A card is drawn from a pack of 52. Find the probability of **3**

(i) an ace

(ii) a heart

(iii) a face card (J, Q, K).

OR

A bag contains 5 red, 3 blue and 2 green balls. Find the probability of drawing **3**

(i) a red ball

(ii) a ball that is not green

(iii) a yellow ball.

18. The probability that a randomly chosen bulb from a batch is defective is 0.04. How many defective bulbs are expected in 1500 bulbs? **3**

OR

A die was rolled 120 times: 1 → 18, 2 → 22, 3 → 19, 4 → 21, 5 → 24, 6 → 16. Find the experimental probability of **3**

(i) getting 5

(ii) getting an even number, and compare

(i) with $\frac{1}{6}$.

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** A survey of 50 students about favourite fruit gives: mango 20, apple 15, banana 10, grapes 5. **5**

(i) Find the probability that a randomly chosen student likes mango.

(ii) Find the probability for each of the other fruits and check that all four add to 1.

(iii) For a school of 1500 students, estimate the number of mangoes and bananas needed.

(iv) How would a better estimate be obtained?

OR

Case study: Two dice are rolled together. **5**

(i) Write how many outcomes the sample space has.

(ii) Find the probability of a sum of 7.

(iii) Find the probability of a double.

(iv) Find the probability that the sum is less than 5.

20. **Case study:** A well-shuffled pack of 52 cards is used. Find the probability of drawing **5**

(i) a king

(ii) a red card

(iii) a face card

(iv) neither a heart nor a diamond

(v) the queen of hearts.

OR

Case study: A bag has 5 red, 3 blue and 2 green marbles. **5**

(i) Find $P(\text{red})$, $P(\text{blue})$, $P(\text{green})$ and show they add to 1.

(ii) Find the probability that the marble is not blue.

(iii) 5 more green marbles are added; find the new $P(\text{green})$.

END OF QUESTION PAPER — Total: 40 Marks

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